

Final Project Assignment: Model Selection

Supervised Learning Module - 2025-3

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Overall Aim:

This assignment aims to assess your proficiency in data analysis, machine learning techniques, and model evaluation. You will work with various synthetic datasets generated using scikit-learn to explore different data scenarios and evaluate the performance of various supervised learning algorithms. Lastly, you will discuss possible avenues of follow-ups for a selected paper from the reading list.

Tasks:

Scenario 1: Binary vs. Multiclass Classification (20 marks)

Aim: Understand the impact of class distribution on model performance in classification tasks.

Dataset Type: Blobs

- **Dataset Generation:** Create and visualize three datasets: one with two classes, one with five classes, and one with twenty classes. In each case, split the data into training, development, and testing sets.
- **Model Selection:** Consider classifiers like k-NN, Decision Tree, SVM, and MLP. Perform hyperparameter tuning (e.g., k, depth, learning rate, epochs, number of layers/neurons) using grid search or random search on the training and development sets for each dataset.
- **Comparison and Analysis:** Compare the performance of different classifiers on the testing set using appropriate metrics (accuracy, precision, recall, F1-score, confusion matrix). Discuss the impact of class distribution on model performance. Visualize the decision boundaries of the classifiers to understand how they separate the classes.

Scenario 2: Big vs. Small Data (10 marks)

Aim: Understand the impact of data availability on model performance.

Dataset Type: Gaussian Quartiles

- **Dataset Generation:** Generate five datasets with four labels and varying sample sizes: 10^2 , 10^3 , 10^4 , 10^5 , and 10^6 . In each case, split the data into training, development, and testing sets.
- **Model Selection:** Consider classifiers like k-NN, Decision Tree, SVM, and MLP. Perform hyperparameter tuning (e.g., k, depth, learning rate, epochs, number of layers/neurons) using grid search on the training and development sets of each dataset.
- **Comparison and Analysis:** Compare the performance of different models on each dataset and discuss the impact of data availability on model performance. Identify which models are better suited for each case and why. Support your arguments with nice visualizations like bar plots, heatmaps, radar plots (let your imagination fly!).

Scenario 3: Clean vs. Noisy Labels (10 marks)

Aim: Understand the impact of noisy labels on model performance.

Dataset Type: Moons

- **Dataset Generation:** Create a binary dataset with two informative features and a sample size of 1000 examples. Select some random samples and artificially corrupt their labels by swapping from positive to negative class or vice versa (simulating noise induction). Use different corruption rates to generate different datasets: 0% (clean dataset), 5%, 10%, 20%, or 30% of the sample size. In each case, split the data into training, development, and testing sets.
- **Model Selection:** Consider classifiers like k-NN, Decision Tree, SVM, and MLP. Perform hyperparameter tuning (e.g., k, depth, learning rate, epochs, number of layers/neurons) using grid search or random search on the training and development sets for each dataset.
- **Comparison and Analysis:** Compare the performance of different models on each dataset and identify the model tolerance to noise. Discuss which models are more robust to noisy labels and why. Support your arguments with nice visualizations like bar plots, heatmaps, radar plots (let your imagination fly!).

Task 4: Discussion of our recent work on LLMs (10 marks)

Aim: Reading and discussing ideas for further research on one selected paper from the following reading list:

From Touch-Tone Menus to Natural Language Intent Routing: An LLM-powered Approach (WEA2025)

[Inprint...](#)

New Kid in the Classroom: Exploring Student Perceptions of AI Coding Assistants (19CCC, 2025)

<https://arxiv.org/abs/2507.22900v2>

Zero-shot spam email classification using pre-trained large language models (2024)

<https://arxiv.org/pdf/2405.15936?>

Using BERT encoding to tackle the mad-lib attack in SMS spam detection (2021)

<https://arxiv.org/pdf/2107.06400>

Submission:

Submit by **September 10th/2025, 5pm**, a PDF version of Jupyter Notebook (Colab) containing your code, explanations, comments, visualizations, performance results and paper discussion onto the given shared folder.

Additional Notes:

- Feel free to experiment with different parameters and algorithms to explore the capabilities on synthetic datasets.
- Use appealing visualization techniques to illustrate your findings.
- Discuss the challenges and limitations encountered during the project.
- You may work in groups of up to 3 people if you believe it would help you manage the workload more effectively.